MUHAMMAD ROSHAAN

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OBJECTIVE

Computer Science 3rd year student with hands-on experience in backend development and data analytics. Skilled in FastAPI, Python, SQL, MySQL/MongoDB, and Power BI. Seeking an internship or junior role to contribute to backend or data-driven projects while gaining industry experience.

EDUCATION

Sir Syed University of Engineering and Technology **Sep 2023–July 2027**
BS in Computer Science | 2023 – Present (**CGPA**:3.81/4.00) ***Karachi***
Courses: Python, SQL, Data Structures, Database Management, Data Analytics, DBMS, OS

TECHNICAL SKILLS

Languages & Tools: Python, SQL, C++, C#, Excel , FastAPI , MongoDB , Supabase
Libraries & Frameworks: Pandas, Matplotlib, NumPy
BI & Visualization Tools: Power BI, DAX, Jupyter Notebook
Core Skills: Data Cleaning, Wrangling, ETL, Dashboarding, Reporting

PRACTICAL EXPERIENCE AND PROJECTS

SeatVault (Concurrency Engine) - Python (FastAPI) + Redis + Celery + React

- Engineered a high-concurrency ticketing system using Redis distributed locks and FastAPI to prevent double-booking and race conditions during flash sales.
- Integrated Celery for asynchronous background processing and built a React frontend with real-time polling and JWT authentication.

BooknScore - Flutter | Supabase | SQLite

- Developed an offline-first cricket scoring mobile app with real-time Supabase cloud synchronization.
- Engineered a resilient SQLite scoring engine to prevent data loss during network interruptions.

Book Manager API – Python (FastAPI) + SQLAlchemy + MySQL

- Built a complete RESTful API for book inventory management, implementing CRUD operations with FastAPI, SQLAlchemy ORM, and MySQL.

FlashSale Concurrency Engine – Python (FastAPI) + MySQL

- Designed a high-concurrency backend system to eliminate Race Conditions in inventory management. Implemented Pessimistic Locking via SQLAlchemy transactions to prevent overselling during simultaneous requests.
- Validated data integrity using a multi-threaded stress-test script simulating heavy traffic loads.

Netflix Data Analysis – Python (Pandas + Matplotlib)

- Explored Netflix dataset using Pandas and Matplotlib to uncover patterns in content release, genre popularity, and geographic distribution. Generated visual insights to highlight trends.

LANGUAGES

English: Fluent | Urdu: Native

CERTIFICATIONS AND COURSES

♣ **Introduction to MongoDB in Python** | DataCamp (Nov 2025)
A 3-hour course covering MongoDB fundamentals, NoSQL concepts, and core database operations in Python.

♣ **FastAPI (Python)** | Udemy | July - Oct 2025
Learned to build and deploy high-performance RESTful APIs with FastAPI, including database integration, data validation, and modular application design (21.5 hours).